

1. Explain the following terms: hit rate, miss rate, hit time, miss penalty.

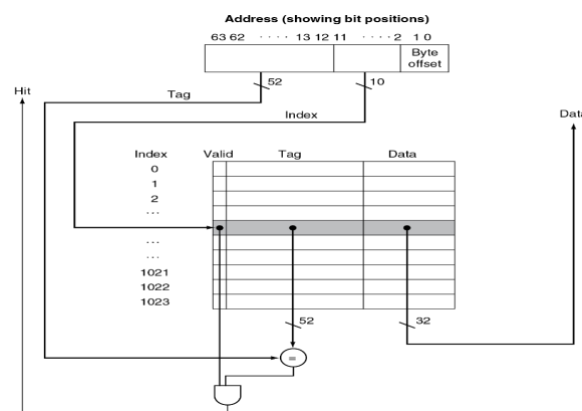
Hit Rate: The percentage of accesses that result in cache hits is known as the hit rate or hit ratio of the cache. When the cache client (a CPU, web browser, operating system) needs to access data presumed to exist in the backing store, it first checks the cache. If an entry can be found with a tag matching that of the desired data, the data in the entry is used instead. This situation is known as a cache hit.

Miss Rate: The fraction or percentage of accesses that result in a miss is called the miss rate. $\text{hit rate} + \text{miss rate} = 1.0$ (100%) The difference between lower-level access time and cache access time is called the miss penalty. Effective access time is a standard effective average.

Hit Time: Hit time is the time to access the upper level of the memory hierarchy, which includes the time needed to determine whether the access is a hit or a miss.

Miss Penalty: The fraction or percentage of accesses that result in a miss is called the miss rate. The difference between lower-level access time and cache access time is called the miss penalty. Effective access time is a standard effective average.

2. Explain the following figure for a one-word per block cache. Explain how “hit” occurs and the data are read.



Answer:

In a 64-bit address space divided into word (2 bits), index (10 bits), and tag (52 bits), the index identifies one of 1024 blocks in the cache. A match between the tag value and the requested address's tag bits signals a cache hit, enabling quick data retrieval. The valid bit (V) indicates if the data in a block is valid. Mismatch in tag leads to a cache miss, prompting data retrieval from main memory. This segmentation optimizes cache access by efficiently organizing and locating data based on their addresses.

3. How many total bits are required for a direct-mapped cache with 2^{12} blocks (12-bit index), two-word in each block, assuming 64-bit address? Compare this total cache size with the 2^{13} words for storage of data as given above.

Hint: n (cache size) and m (one bit for selecting two-word block) is given, and then derive tag size (two bits for byte offset). Then, find the total bits for address/data and compare that with pure data bits. What is the ratio?

Answer:

$$\text{Number of Blocks} = 2^{12} = 2^n$$

$$\text{Block size} = 2 \text{ word} = 2^1 = 2^m$$

$$\text{Total bits} = 64$$

$$2^n * (2^m * 32 + (64 - n - m - 2) + 1) = 2^{12} * (2 * 32 + (64 - 12 - 3 - 2) + 1) = 1253376$$

$$\text{Ratio} = 1253376 / 2^{13} = 153$$

4. Below is a list of memory address references in HEX: 0x03, 0xb3, 0x03, 0x03, 0xbf, 0xcf, 0xbe, 0x0e, 0x05, 0x25, 0x0f, 0x0e
- a. For each of these references, in a table, identify the binary word address, the tag, and the index given a direct-mapped cache with **16 one-word** blocks (thus lower 4 bits are index and we have 32-bit data). Also list whether each reference is a hit or a miss, assuming the cache is initially empty. Ignore byte offset bits for simplicity.

Answer:

Word Address	Binary Address	Tag	Index	Miss/Hit
0x03	0000 0011	0000	0011	Miss
0xb3	1011 0011	1011	0011	Miss
0x03	0000 0011	0000	0011	Miss
0x03	0000 0011	0000	0011	Hit
0xbf	1011 1111	1011	1111	Miss
0xcf	1100 1111	1100	1111	Miss
0xbe	1011 1110	1011	1110	Miss
0x0e	0000 1110	0000	1110	Miss
0x05	0000 0101	0000	0101	Miss
0x25	0000 0101	0010	0101	Miss
0x0f	0000 1111	0000	1111	Miss
0x0e	0000 1110	0000	1110	Hit

- b. For each of these references, in a table, identify the binary word address, the tag, the index, and the offset given a direct-mapped cache with **two word** blocks ($32 \times 2 = 64$ -bit data) and a total size of **eight blocks**. Also list if each reference is a hit or a miss, assuming the cache is initially empty.
Consider one bit for word offset and ignore byte offset bits for simplicity.

Answer:

Word Address	Binary Address	Tag	Index	Offset	Miss/Hit
0x03	0000 0011	0000	001	1	Miss
0xb3	1011 0011	1011	001	1	Miss
0x03	0000 0011	0000	001	1	Miss
0x03	0000 0011	0000	001	1	Hit
0xbf	1011 1111	1011	111	1	Miss
0xcf	1100 1111	1100	111	1	Miss
0xbe	1011 1110	1011	111	0	Miss
0xoe	0000 1110	0000	111	0	Miss
0x05	0000 0101	0000	010	1	Miss
0x25	0000 0101	0010	010	1	Miss
0x0f	0000 1111	0000	111	1	Hit
0x0e	0000 1110	0000	111	0	Hit

5. Assume the miss rate of an instruction cache is 4% and the miss rate of the data cache is 5%. If a processor has a CPI of 3 without any memory stalls, and the miss penalty is 50 cycles for all misses, determine how much faster a processor would run with a perfect cache that never missed. Assume the frequency of all loads and stores is 44%.

Answer:

Miss rate of instruction cache = 0.04

Miss rate of data cache = 0.05

Average CPI = 3 cycles

Miss penalty = 50 cycles

Frequency of all loads and stores = 44%

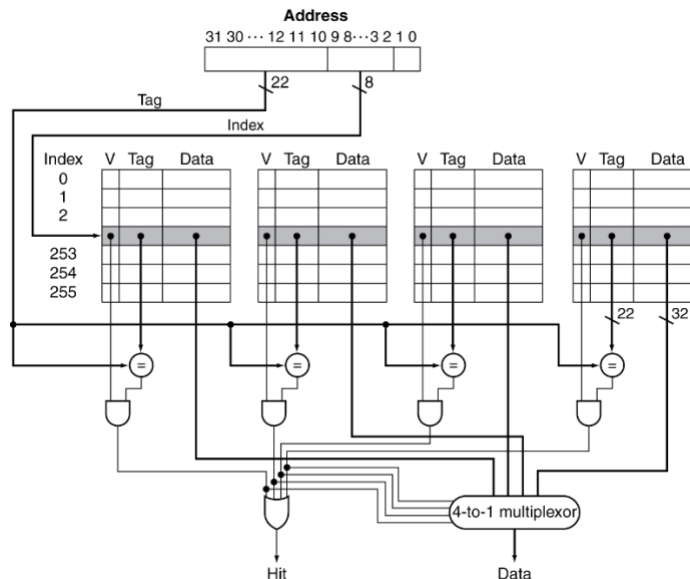
Instruction cache = $0.04 \times 50 = 2$

Data cache = $0.44 \times 0.05 \times 50 = 1.1$

Actual CPI = $2 + 1.1 + 3 = 6.1$

Ratio of CPU execution time = $6.1/3 = 2.03$ times faster

6. (1 point) Explain the set associative architecture below:



Answer:

Three parts make up an address in a set associated mapping: tag bits, set offset, and block offset. The address in this case has 32 bits, of which 22 are tag bits, 8 are set offset bits, and 2 are byte offset bits. A comparator is used to compare the tag bits in the address with the tag bits in the cache memory. The number of bits used for the tag equals the comparator's size. Since the tag in this case has 22 bits, the comparator that is utilised will likewise have 22 bits. With only 8 bits in the set offset field, the cache contains 28 sets. Number of lines or blocks in a cache equals number of comparators in a cache. Since each cache in this image only displays one comparator, we can conclude that each set contains a single block. The cache is displayed as a table with the three properties of data, tag, and V. Here, V denotes validity. It indicates whether the cache is being used. The related tag value is contained in the tag field. The data portion of the block is indicated by data offcoure.

Therefore, the tag bits are compared using comparators from each cache when the CPU constructs an address. Should they be identical, the comparator's output will be 1. An AND gate, to which the valid bit is also an input, is connected to this output.

1. The value must first be written back to the main memory if V=0, which indicates that the data has been consumed. As a result, in that scenario, the AND gate's output will be 0, and the OR gate will likewise produce 0, signifying a miss.
2. However, if V=1 (data that is not being used) then the AND gate will yield 1 and the OR gate's output will also yield 1 indicating a hit. And as a result, the MUX will produce the appropriate data as its output.

7. (2 points) For the following references (in decimal) and a 4-block cache, fill in the following two tables: 0, 1, 2, 3, 6

Hint: For each of the tables below, you should get 5 misses.

Answer:

Direct Mapped						
Block address	Cache index	Hit/Miss	Cache content after access			
			0	1	2	3
0000	00	Miss	MEM[0]			
0001	01	Miss	MEM[0]	MEM[1]		
0010	10	Miss	MEM[0]	MEM[1]	MEM[2]	
0011	11	Miss	MEM[0]	MEM[1]	MEM[2]	MEM[3]
0110	10	Miss	MEM[0]	MEM[1]	MEM[6]	MEM[3]

2 -Way Set Associative						
Block address	Cache index	Hit/Miss	Cache content after access			
			Set 0		Set 1	
0000	0	Miss	MEM[0]			
0001	1	Miss	MEM[0]		MEM[1]	
0010	0	Miss	MEM[0]	MEM[3]	MEM[1]	
0011	1	Miss	MEM[0]	MEM[3]	MEM[1]	MEM[3]
0110	0	Miss	MEM[6]	MEM[3]	MEM[1]	MEM[3]